

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0518

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- (a) Clearly interpret the scenario and explain the cause of low contrast.
- (b) Identify all key issues affecting image visibility.
- (c) Apply an appropriate enhancement technique to address the problem.
- (d) Justify your choice with logical reasoning.
- (e) Present your explanation in a clear and structured manner.

Low Contrast Image

(a) Interpretation:

The grayscale image has very small intensity differences between the object and the background, making it dull and difficult to distinguish.

(b) Key issues:

- Poor visibility
- Reduced dynamic range
- Loss of important details
- Difficult segmentation

(c) Enhancement technique:

Histogram Equalization.

(d) Justification:

Histogram equalization spreads intensity values across the entire range, improving contrast and visibility.

(e) Conclusion:

The technique enhances object-background separation and improves image quality.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

- (a) Interpret the scenario and explain the impact of uneven illumination.
- (b) Identify key factors causing intensity variation.
- (c) Apply a suitable enhancement method to normalize illumination.
- (d) Justify your selected approach with reasoning.
- (e) Provide a clear and well-structured explanation.

Uneven Illumination

(a) Interpretation:

The image contains bright and dark regions due to non-uniform lighting.

(b) Key issues:

- Intensity variation
- Shadow effects
- Loss of details

(c) Enhancement technique:

Adaptive Histogram Equalization (CLAHE).

(d) Justification:

CLAHE improves local contrast and normalizes illumination.

(e) Conclusion:

The method balances brightness and enhances visibility.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Explain the scenario and characteristics of this noise.
- (b) Identify key issues impacting image quality.
- (c) Apply an appropriate filtering technique.
- (d) Justify why this method is more suitable than alternatives.
- (e) Present your response clearly and logically.

Salt-and-Pepper Noise

(a) Interpretation:

Random black and white pixels affect document readability.

(b) Key issues:

- Noise artifacts
- Reduced readability
- Loss of information

(c) Filtering technique:

Median Filter.

(d) Justification:

Median filtering removes impulse noise while preserving edges better than averaging filters.

(e) Conclusion:

The document becomes cleaner and more readable.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- (a) Interpret the scenario and explain why edges are blurred.
- (b) Identify key issues related to smoothing and edge loss.
- (c) Apply an appropriate technique to restore edges.
- (d) Justify your decision with clear reasoning.
- (e) Structure your explanation clearly.

Blurred Edges

(a) Interpretation:

Excessive smoothing has removed important edge details.

(b) Key issues:

- Edge loss
- Reduced sharpness
- Poor object boundaries

(c) Technique:

Unsharp Masking.

(d) Justification:

It enhances edge information and restores sharpness.

(e) Conclusion:

Object boundaries become clearer.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- (a) Interpret the scenario and describe high-frequency noise.
- (b) Identify the key issues affecting image clarity.
- (c) Apply an appropriate frequency domain filtering method.
- (d) Justify the choice of filter with reasoning.
- (e) Present your answer in a clear and organized manner.

High-Frequency Noise

(a) Interpretation:

Fine-grained noise affects satellite image quality.

(b) Key issues:

- Reduced clarity
- Hidden features
- Distorted texture

(c) Technique:

Low-pass frequency filter.

(d) Justification:

It suppresses high-frequency noise while preserving useful information.

(e) Conclusion:

The image becomes smoother and easier to analyze.